SoulHeal

Student Mental Wellness Platform

Project Report

Team – 11

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1. Introduction

Students often face a wide range of challenges during their educational journey — including stress, anxiety, academic pressure, loneliness, and various emotional difficulties. Contributing factors such as examinations, project deadlines, peer pressure, family expectations, and career uncertainty can significantly affect a student's mental well-being.

Despite the growing need for support, many students hesitate to seek professional help due to fear, social stigma, lack of awareness, or limited access to counseling services. This gap can lead to worsening emotional conditions that affect academic performance and quality of life.

SoulHeal is a web-based mental wellness platform built to bridge this gap. It provides students with a safe, confidential, and private digital space where they can track their mood, write reflective journals, access stress-relief therapy tools, and connect with counselors when needed — all from a single platform.

2. Project Description

SoulHeal is a web-based Student Mental Health Platform developed using the MERN Stack (MongoDB, Express.js, React.js, Node.js). The main purpose of this platform is to support students in maintaining their emotional and mental well-being through digital self-help tools, counseling support, and wellness resources.

The platform allows students to create secure accounts and access various mental health features. Students can use mood tracking tools to record how they feel every day and identify emotional patterns over time. They can also take self-assessment tests related to stress, anxiety, emotional balance, focus, and overall well-being. Based on the results, the platform provides suggestions, mindfulness exercises, and motivational resources.

SoulHeal also allows students to connect with counselors or mentors through appointment booking, chat support, or virtual sessions. Students can request guidance confidentially without fear of judgment. Counselors can monitor student requests, review self-assessment results, and provide appropriate support or recommendations.

The system maintains privacy and security by ensuring that student information is protected and only accessible to authorized users. Role-based access control is implemented so that students, counselors, and administrators have different levels of access — students can only view their own records, counselors can access only assigned student requests, and administrators can manage overall platform activities.

The platform also provides wellness resources such as guided meditation, breathing exercises, motivational videos, mental health articles, stress management tips, and emergency helpline information. These resources help students manage emotional challenges independently before problems become severe.

3. Scenario-Based Case Study

3.1 The Student: Priya

Priya is a college student who feels stressed because of exams, project deadlines, and placement preparation. She does not feel comfortable sharing her emotions with friends or family. Instead of ignoring her feelings, she uses SoulHeal to manage her stress in a safe and private environment.

3.2 Step-by-Step User Journey

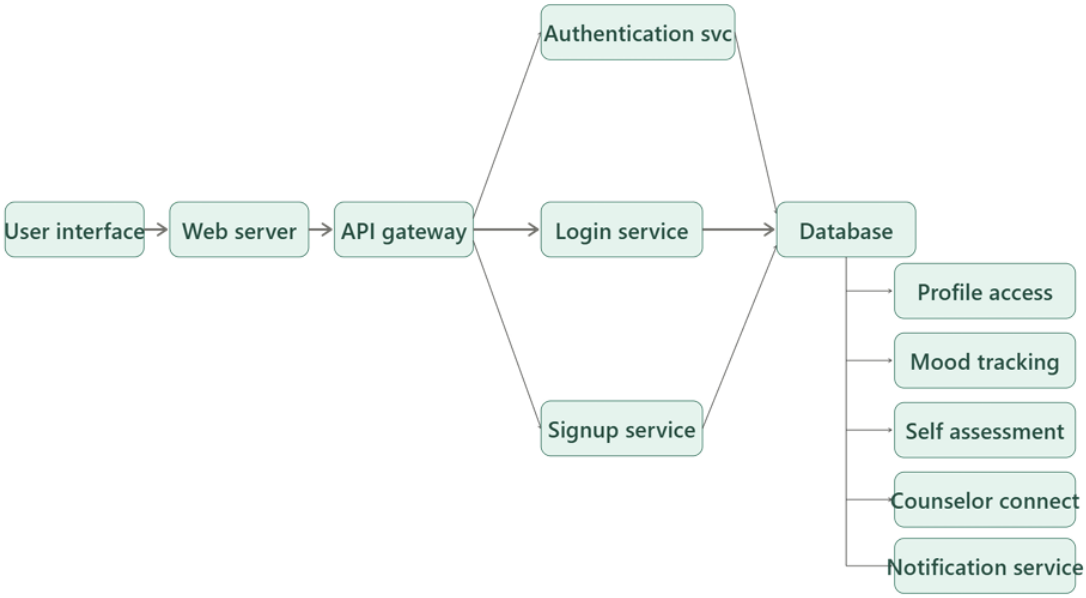
#	Stage	What Priya Does
1	**User Registration / Login**	Priya creates a secure account and logs into the SoulHeal platform.
2	**Mood Tracking**	She records her mood every day and notices she feels anxious frequently before exams.
3	**Self-Assessment**	Priya takes a stress assessment quiz provided by the platform to measure her current state.
4	**Resource Access**	Based on the results, SoulHeal suggests meditation exercises and stress management tips.
5	**Counselor Support**	Priya books an appointment with a counselor for additional personalized guidance.
6	**Secure Communication**	The counselor reviews her assessment and provides support through chat and virtual meetings.
7	**Progress Monitoring**	Priya continues using the mood tracker and notices improvement in her emotional well-being over time.

4. Technical Architecture

The SoulHeal platform is built on a layered, modular architecture. Each layer is responsible for a specific concern — from user interface rendering to data persistence — ensuring clean separation and scalability.

4.1 Architecture Components

Component	Technology	Responsibility
Frontend Application	React.js, HTML, CSS	Renders all UI pages — Dashboard, Journal, Therapy, Profile. Communicates with backend via Axios.
Backend Server	Node.js, Express.js	Handles all API routes, business logic, middleware, and HTTP request/response lifecycle.
Authentication Module	JWT, bcrypt	Manages user registration, login, password hashing, token generation and protected route verification.
Mood Tracking Module	Express Routes + MongoDB	Stores daily emotional entries. Supports full CRUD for journal records per user.
Assessment Module	React + MongoDB	Serves self-assessment quizzes, scores results, and stores them for counselor review.
Counseling Module	Express + MongoDB	Enables appointment booking and communication between students and counselors.
Notification Service	Nodemailer (optional)	Sends email reminders, appointment confirmations, and motivational alerts.
Database	MongoDB + Mongoose	Stores all collections: Users, Journals, Assessments, Appointments, and Resources.
Admin Panel	React + Protected Routes	Lets administrators manage users, counselors, resources, and view platform reports.


Authentication svc
User interface
Web server
API gateway
Login service
Database
Profile access
Mood tracking
Signup service
Self assessment
Counselor connect
Notification service

5. ER Diagram & Relationships

SoulHeal uses MongoDB as its database. Although MongoDB is a NoSQL document-based database, the collections are designed with clear relationships to reflect real-world associations between entities. The following describes the core collections and their relationships.

5.1 Core Collections (Entities)

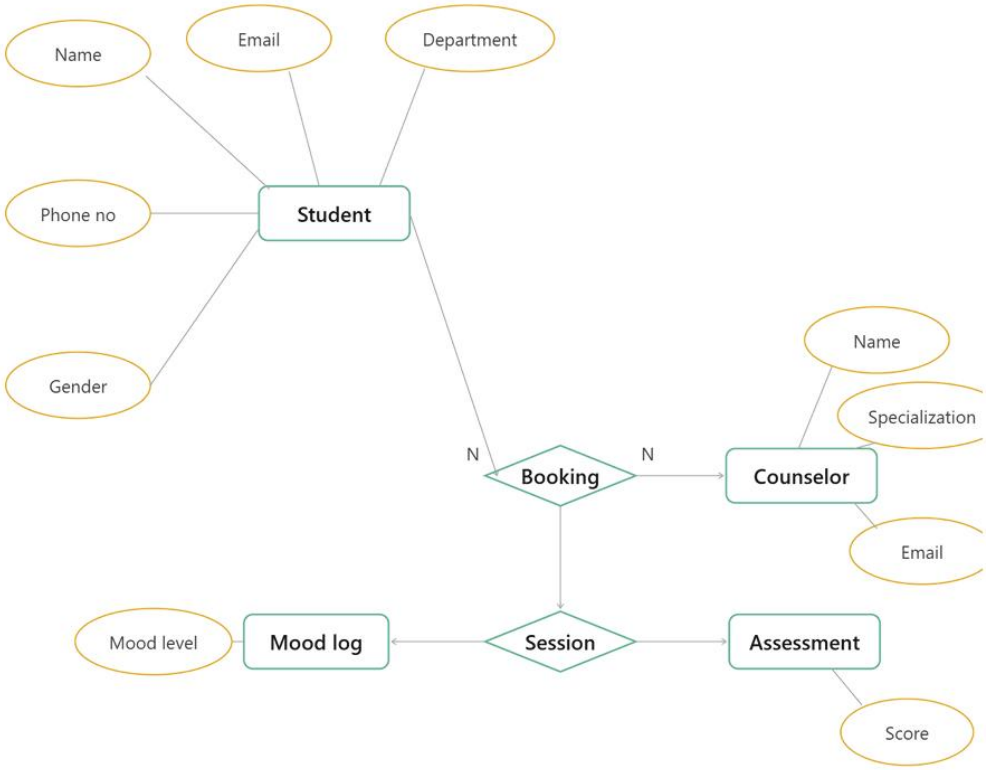
Collection	Key Fields	Description
User	_id, name, email, password (hashed), role, createdAt	Stores all registered users. The role field distinguishes between student, counselor, and admin.
Journal	_id, userId (ref: User), mood, notes, date, createdAt	Each journal entry belongs to one user. Stores daily mood type and personal notes.
Assessment	_id, userId (ref: User), type, score, responses[], date	Stores self-assessment quiz results for each student. Linked to the user by userId.
Appointment	_id, studentId (ref: User), counselorId (ref: User), date, status, notes	Links a student to a counselor for a scheduled session. Tracks appointment status.
Resource	_id, title, type, content, addedBy (ref: User), createdAt	Wellness resources (articles, videos, exercises) managed by the admin.

5.2 Entity Relationships

Relationship	Type	Description
User → Journal	One-to-Many	One user can have many journal entries. Each journal entry belongs to exactly one user.
User → Assessment	One-to-Many	One user can complete many assessments over time. Each assessment belongs to one user.
User → Appointment	One-to-Many	A student user can book multiple appointments. Each appointment references both a student and a counselor.
Counselor → Appointment	One-to-Many	A counselor can be assigned to multiple appointments from different students.
Admin → Resource	One-to-Many	An admin user can add multiple wellness resources to the platform.

5.3 ER Diagram

The relationships between entities are summarized below:



6. Project Flow

The following describes the complete end-to-end operational flow of the SoulHeal platform — from a student first visiting the site to logging out after a productive session.

6.1 Student Flow

Step	Action	Module Involved
01	Student registers with name, email, and password.	Authentication Module
02	Password is hashed with bcrypt and saved to MongoDB.	Auth + Database
03	Student logs in — backend validates credentials and issues JWT token.	Authentication Module
04	Student lands on the Dashboard — streak and quick actions are displayed.	Dashboard
05	Student records daily mood — selects mood type and writes personal notes.	Mood Tracking Module
06	Journal entry is saved to MongoDB via POST /api/journal (JWT protected).	Journal + Database
07	Student takes a self-assessment quiz for stress or anxiety.	Assessment Module
08	Platform scores the quiz and suggests relevant wellness resources.	Assessment + Resources
09	Student browses wellness resources — reads articles or does breathing exercises.	Wellness Resources
10	Student books an appointment with a counselor.	Counseling Module
11	Counselor receives the request and reviews mood + assessment data.	Counseling Module
12	Counselor communicates guidance through chat or virtual session.	Chat / Video Module
13	Student continues daily check-ins — emotional trends visible on profile.	Profile + Tracking
14	Student logs out — JWT removed from localStorage, session ends securely.	Authentication Module

6.2 Counselor Flow

Step	Action
01	Counselor logs in with their assigned credentials.
02	Counselor views list of student appointment requests assigned to them.
03	Counselor reviews mood tracking history and assessment results of the student.
04	Counselor accepts the appointment and communicates via chat or virtual session.
05	Counselor updates appointment status and adds session notes to the record.

6.3 Admin Flow

Step	Action
01	Admin logs in via the admin panel with elevated access credentials.
02	Admin manages user accounts — can add, update, or deactivate student/counselor profiles.
03	Admin adds or updates wellness resources (articles, videos, exercises) on the platform.
04	Admin monitors appointment activity and overall platform usage statistics.
05	Admin generates reports on mental health trends, resource usage, and user engagement.

7. Project Explanation

7.1 What Was Built

SoulHeal is a mental wellness web application developed using the MERN stack (MongoDB, Express.js, React.js, Node.js). The application enables users to track their daily mood, maintain a private journal, and use built-in therapy tools such as breathing exercises and sound therapy. The platform is designed with a calm, app-like aesthetic inspired by wellness applications like Headspace, prioritizing simplicity, emotional comfort, and ease of use.

7.2 Why It Was Built

Mental health is a growing concern among students and young adults, yet dedicated tools for daily emotional tracking remain scarce or overly complex. SoulHeal was built to provide:

- A simple and consistent way to track daily emotional states
- Instant access to stress-relief tools without needing professional appointments
- A private, personal space for emotional expression through journaling
- Lightweight therapy features (sound and breathing) for immediate calming

7.3 Problem It Solves

Problem	SoulHeal's Solution
Users don't track emotions regularly	Daily mood journal with quick mood selection and notes
No simple tools for quick stress relief	Built-in sound therapy and guided breathing animation
Lack of personalized mental health apps	Activity streak, profile tracking, and personal dashboard
Fear of judgment when seeking help	Private, secure, and fully confidential personal space

7.4 Main Modules

Module	Description
Authentication	Secure user registration, login, and protected route access using JWT
Dashboard	Shows activity streak, quick action buttons, and overall navigation
Mood Journal (CRUD)	Users can create, read, update, and delete personal mood entries
Therapy Tools	Sound therapy player and animated breathing exercise for relaxation
Profile & Tracking	Displays user info and emotional activity history over time

8. Packages Installed and Their Purpose

The SoulHeal project uses carefully selected npm packages for both the backend and frontend. Each package serves a specific, well-defined purpose in the application.

8.1 Backend Packages

Package	Purpose
express	Creates the Node.js server, defines API routes, and handles all HTTP requests and responses.
mongoose	Connects Node.js with MongoDB and provides schema-based data modeling for User and Journal collections.
cors	Enables Cross-Origin Resource Sharing so the React frontend can communicate with the Express backend.
dotenv	Loads environment variables (database URL, JWT secret) from a .env file, keeping sensitive data secure.
bcrypt	Hashes user passwords before saving to the database, ensuring passwords are never stored in plain text.
jsonwebtoken	Generates and verifies JWT tokens for stateless, secure authentication after login.
nodemon	Automatically restarts the backend server during development whenever code changes are saved.

8.2 Frontend Packages

Package	Purpose
react-router-dom	Handles client-side navigation between pages (Login, Dashboard, Journal, Therapy, Profile).
axios	Makes HTTP API calls from the React frontend to the Express backend for data operations.
react-swipeable	Enables swipe gesture navigation on mobile devices, giving the app a native mobile-like feel.

9. Authentication

Authentication is one of the most critical parts of SoulHeal. Since the platform stores personal emotional data, secure and reliable login is essential to protect user privacy.

9.1 How Login Works — Step by Step

Step	Action
1	User enters their registered email and password on the login page.
2	The frontend sends a POST request to the backend /api/auth/login endpoint via Axios.
3	The backend finds the user record in MongoDB using the provided email.
4	bcrypt compares the entered password against the hashed password stored in the database.
5	If the credentials are correct, the backend generates a JWT token signed with a secret key.
6	The JWT token is sent back to the frontend in the API response.
7	The frontend stores the token in localStorage for persistent login across sessions.
8	All subsequent protected API calls include the token in the Authorization header.

9.2 Why JWT Was Chosen

JSON Web Tokens (JWT) were selected for authentication in SoulHeal for the following reasons:

- Stateless — the server does not need to store session data, making the system scalable
- Secure — tokens are signed with a secret key and expire after a set time
- Industry standard — widely used in modern MERN stack applications
- No session storage needed — reduces server memory usage and complexity
- Easy to use with protected routes — middleware verifies the token before granting access

10. CRUD Operations

CRUD (Create, Read, Update, Delete) forms the backbone of SoulHeal's Mood Journal module. Every journal entry a user creates is fully manageable — they can add new entries, view past ones, edit existing entries, and delete them when needed.

10.1 CRUD in the Journal Module

Operation	HTTP Method	Endpoint	Description
Create	POST	/api/journal	User submits a new mood entry with mood type and written notes.
Read	GET	/api/journal	Fetches all journal entries belonging to the logged-in user.
Update	PUT	/api/journal/:id	User edits the mood type or written content of an existing entry.
Delete	DELETE	/api/journal/:id	User removes an unwanted entry permanently from the database.

10.2 Why CRUD Matters

CRUD operations are the core of backend development and real-world data handling. Implementing CRUD in SoulHeal demonstrates complete mastery of the backend-to-database workflow — from receiving a request on Express, validating the user via JWT middleware, running the appropriate Mongoose operation on MongoDB, and returning the correct response to the frontend.

11. MVC Architecture

SoulHeal follows the Model-View-Controller (MVC) design pattern. This is a standard architectural approach that separates the application into three distinct layers — data, logic, and presentation — making the codebase clean, maintainable, and easy to scale.

11.1 MVC Breakdown in SoulHeal

Layer	Files / Folders	Responsibility
Model	User.js, Journal.js	Defines the structure and schema of data stored in MongoDB using Mongoose.
View	React Components (frontend)	Renders the user interface — Dashboard, Journal, Therapy, Profile pages.
Controller	authController.js, journalController.js	Contains all business logic — handles login, CRUD operations, and data processing.
Routes	authRoutes.js, journalRoutes.js	Defines API endpoint paths and maps them to the correct controller functions.
Middleware	authMiddleware.js	Verifies JWT tokens before granting access to protected API routes.

11.2 Why MVC?

- Clean separation of concerns — each layer has one job and does it well
- Easier debugging — issues in logic, data, or UI can be traced and fixed independently
- Scalability — new features can be added without disrupting existing code
- Industry standard — MVC is used in professional-grade applications worldwide

12. Frontend and Backend Connection

The React frontend and the Node.js/Express backend are two separate applications that communicate through REST API calls. Axios is used on the frontend to make these HTTP requests.

12.1 How They Are Connected

Layer	Technology	Role
Frontend	React.js + Axios	Sends HTTP requests (GET, POST, PUT, DELETE) to the backend API endpoints.
Backend	Node.js + Express.js	Receives requests, processes logic through controllers, and returns JSON responses.
Database	MongoDB + Mongoose	Stores and retrieves all persistent data (users, journal entries).
Auth Layer	JWT in Headers	Every protected request includes the JWT token in the Authorization header.
CORS	cors package	Configured to allow requests from the React dev server to the Express server.

12.2 Data Security in Transit

All API calls that involve user-specific data (fetching journals, updating entries, accessing the profile) are protected. The JWT token is attached to every request header, and the backend middleware validates it before processing. This ensures that users can only access their own data, and unauthorized requests are rejected with a 401 Unauthorized response.

13. UI/UX Design

The user interface of SoulHeal was designed with mental wellness in mind. Visual design choices were intentionally calming, minimal, and intuitive — inspired by leading wellness apps like Headspace and Calm.

13.1 Design Principles

Design Choice	Reason
Calm color palette with soft gradients	Creates a relaxing, non-stressful visual environment suitable for mental health contexts.
Glassmorphism UI style	Gives the app a modern, premium feel with frosted-glass effect cards and panels.
Swipe navigation (react-swipeable)	Provides a mobile-native experience, allowing users to navigate pages with natural gestures.
Mobile-like layout with no long scrolling	Keeps each screen focused on one task, reducing cognitive load and improving usability.
App-like single-page feel	Users navigate instantly without full page reloads, creating a fast and smooth experience.

13.2 Page-by-Page UI Overview

Page	Key UI Elements
Login / Register	Clean centered card, email/password fields, smooth transition to dashboard on login.
Dashboard	Activity streak counter, quick-action buttons, navigation to all modules.
Mood Journal	Mood selector icons, text area for notes, list of past entries with edit/delete options.
Therapy	Audio player for sound therapy, animated breathing circle with inhale/exhale guidance.
Profile	User details display, activity summary, and emotional tracking history.

14. Key Features

SoulHeal is built around a focused set of features that address the everyday mental wellness needs of students. Each feature was designed to be simple, effective, and immediately accessible.

Feature	Description
User Authentication	Secure registration and login with bcrypt password hashing and JWT-based session management.
Dashboard with Streak	Home screen showing activity streak, motivational status, and quick navigation buttons.
Mood Journal (CRUD)	Full create, read, update, delete functionality for personal daily mood entries.
Sound Therapy	Built-in audio player with calming soundscapes to help users relax and de-stress.
Breathing Exercise	Animated guided breathing tool with timed inhale/hold/exhale cycles for anxiety relief.
Profile & Activity Tracking	Displays user info and a summary of emotional activity and journaling history.
Swipe Navigation	Mobile-friendly swipe gestures between pages powered by react-swipeable.
Protected Routes	JWT middleware ensures only authenticated users can access personal data and features.
Secure Data Storage	All user data is stored in MongoDB with hashed passwords and token-protected access.
Responsive Design	App-like layout that works seamlessly across mobile, tablet, and desktop screens.

15. Roles and Responsibilities

SoulHeal implements a user-centric role structure. While the platform is primarily built for students, the backend supports role-based access that can be extended for counselors and administrators.

15.1 Student User)

- Register and log in securely to the platform
- Track mood and emotional states through daily journal entries
- Use therapy tools (sound and breathing) for immediate stress relief
- View personal activity history and emotional progress on the profile page
- Edit and delete personal journal entries at any time

15.2 Counselor / Extended Role)

- Review student mood patterns and journal summaries (with consent)
- Provide professional guidance and emotional support through the platform
- Manage counseling sessions and communication records

15.3 Administrator

- Manage registered users and counselor accounts
- Monitor overall platform usage and activity
- Add and update wellness resources available to users
- Ensure platform security, performance, and data integrity

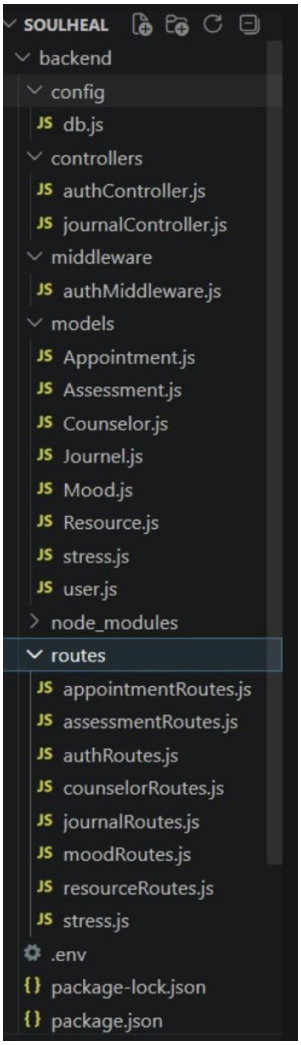
16. Technical Architecture

The SoulHeal platform follows a clean, modular MERN stack architecture with clear separation between client and server layers.

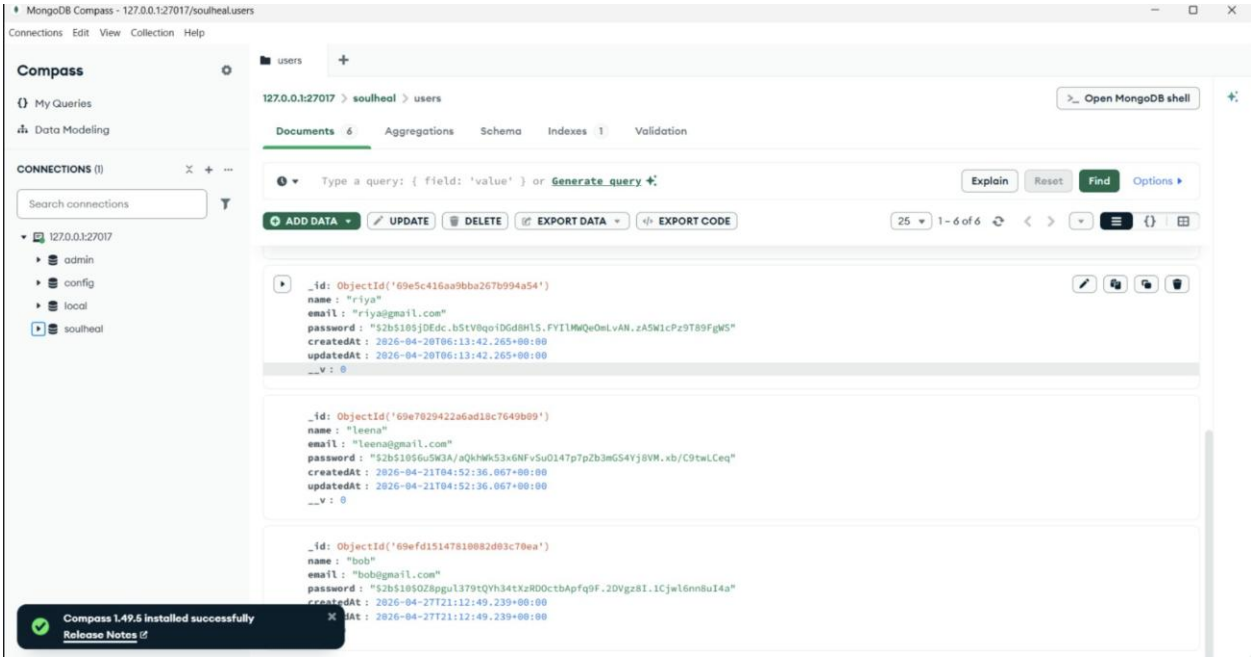
Component	Technology	Description
Frontend Application	React.js	Interactive SPA with routing, state management, and API integration.
Backend Server	Node.js + Express.js	RESTful API server handling all routes, logic, and database operations.
Database	MongoDB + Mongoose	NoSQL database storing users, journal entries, and session metadata.
Authentication	JWT + bcrypt	Token-based auth with hashed passwords and middleware-protected routes.
API Communication	Axios + CORS	Frontend-to-backend communication with cross-origin support.
Real-time (Optional)	Socket.io	Can be added for real-time chat between students and counselors.
Email Alerts (Optional)	Nodemailer	Can be used to send appointment or wellness reminder emails.
Dev Tools	Postman, Git, GitHub	API testing, version control, and team collaboration.

16. Project Setup And Configuration.

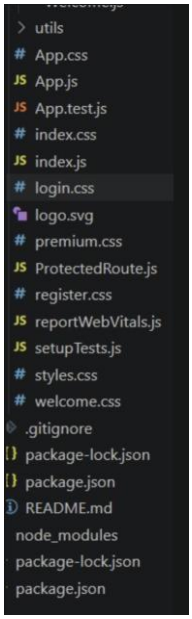
BACKEND DEVELOPMENT



DATABASE DEVELOPMENT:

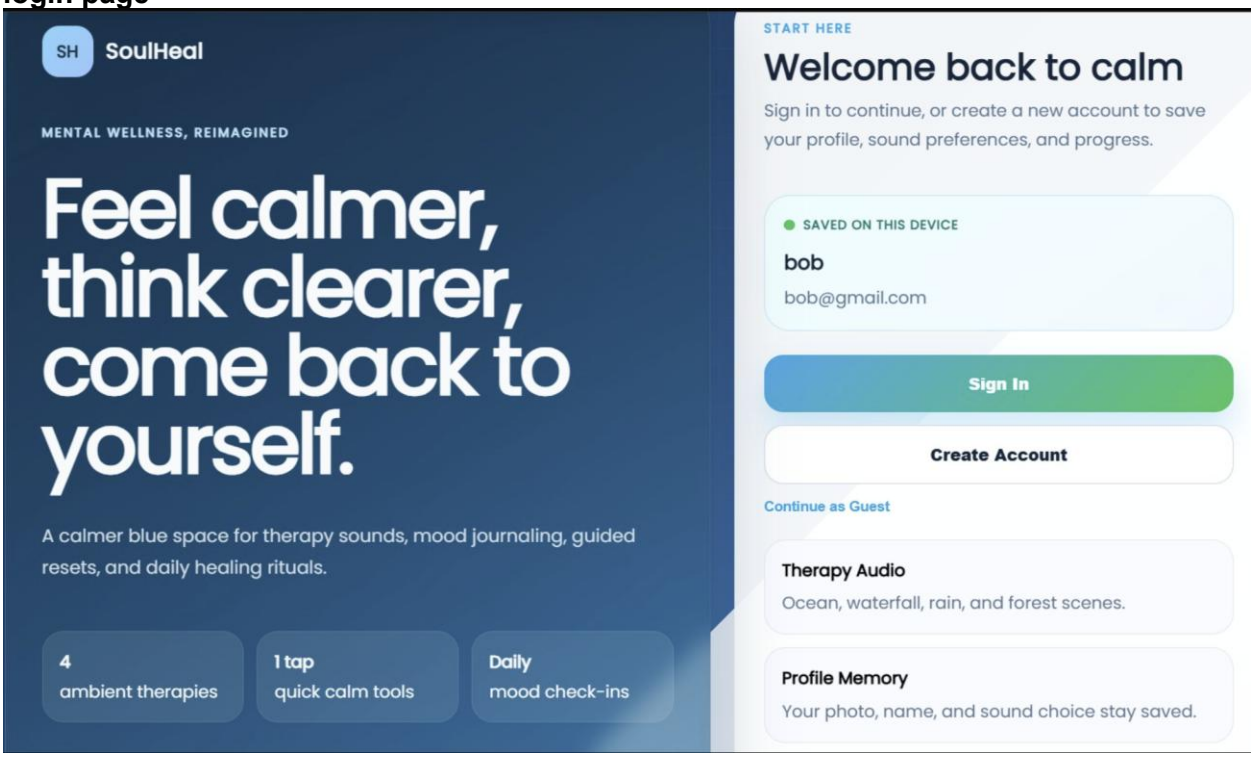


FRONT-END DEVELOPMENT:

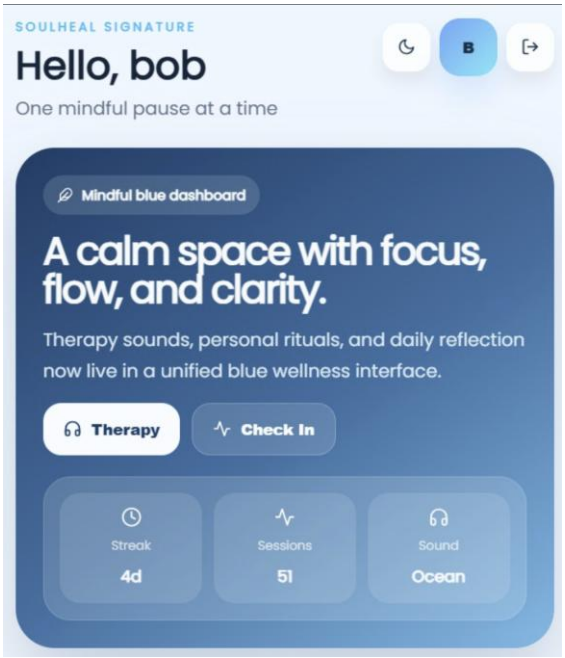


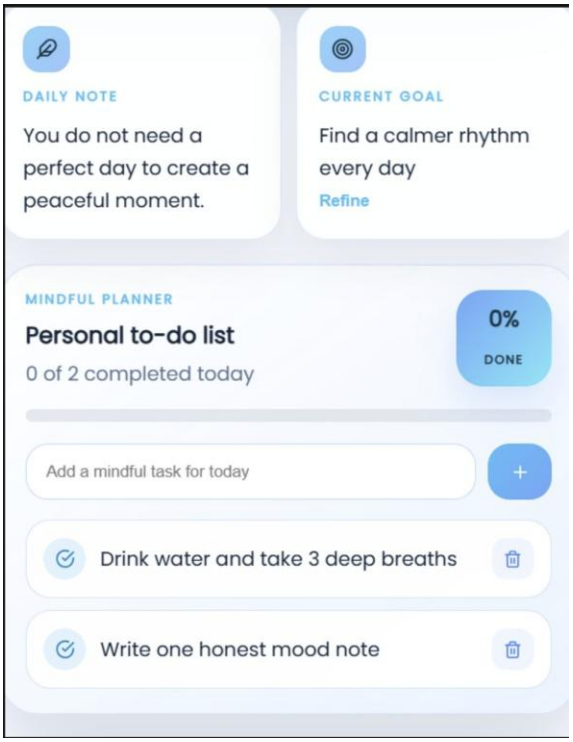
17. Output

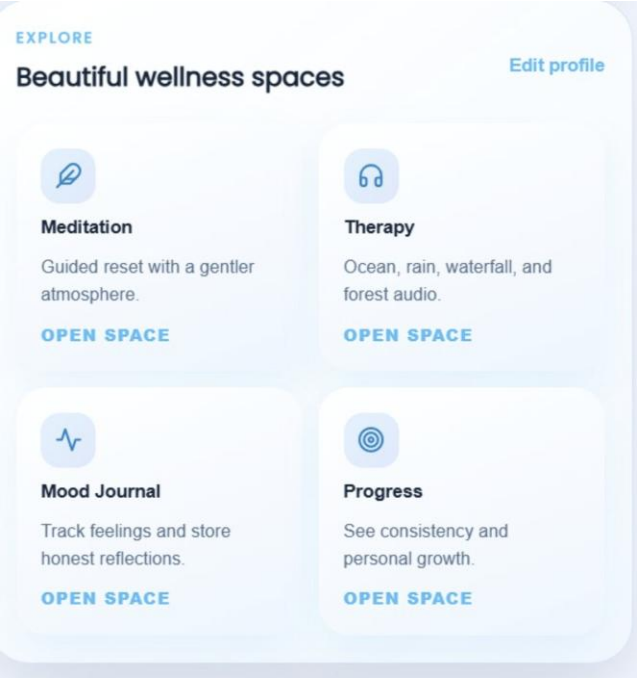
login page



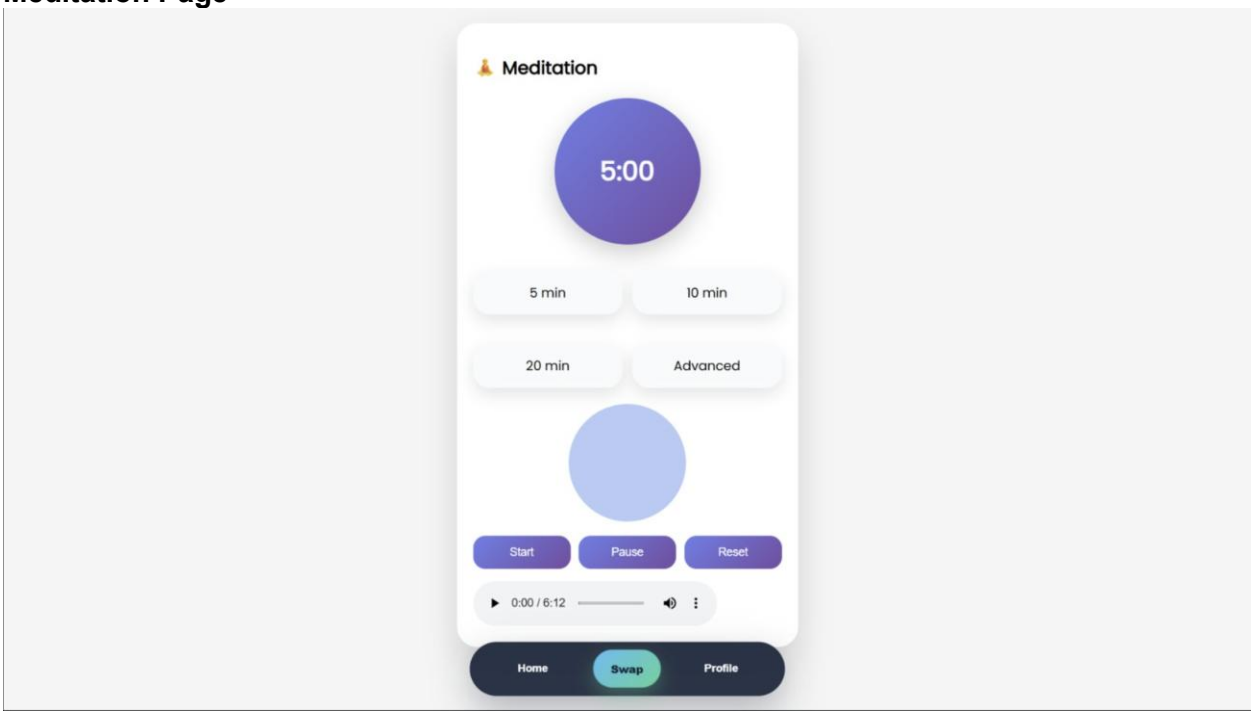
Home page



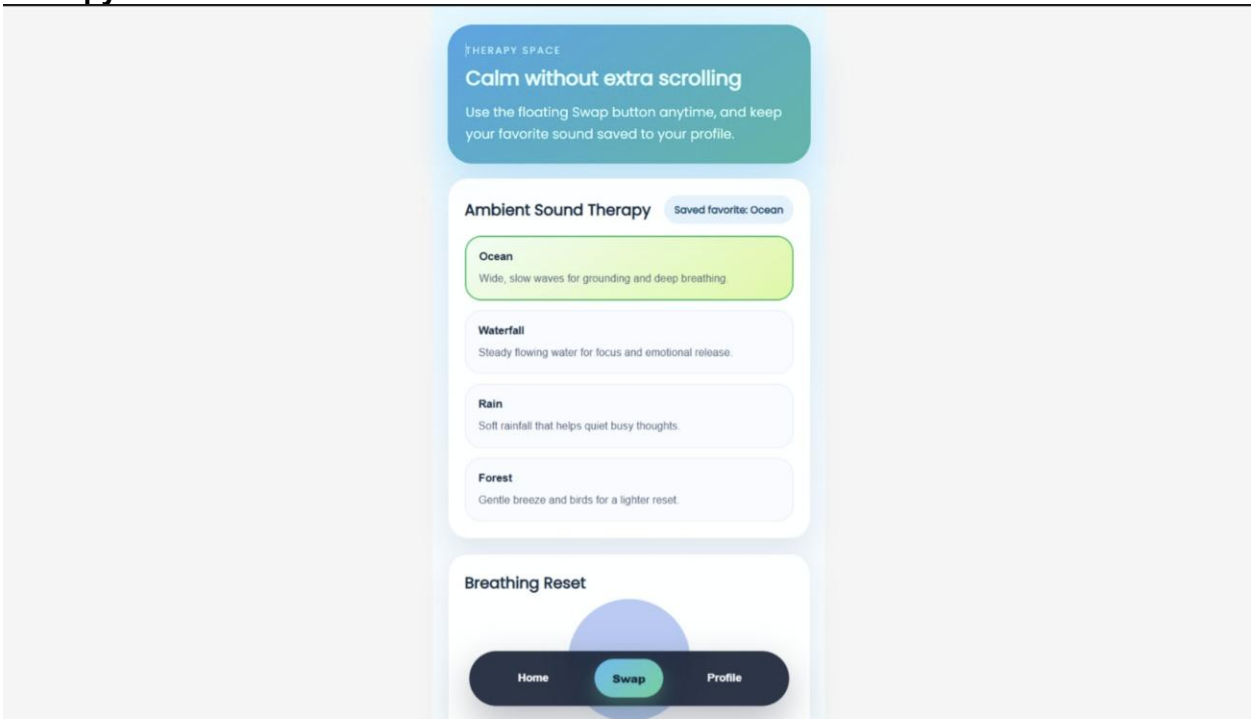




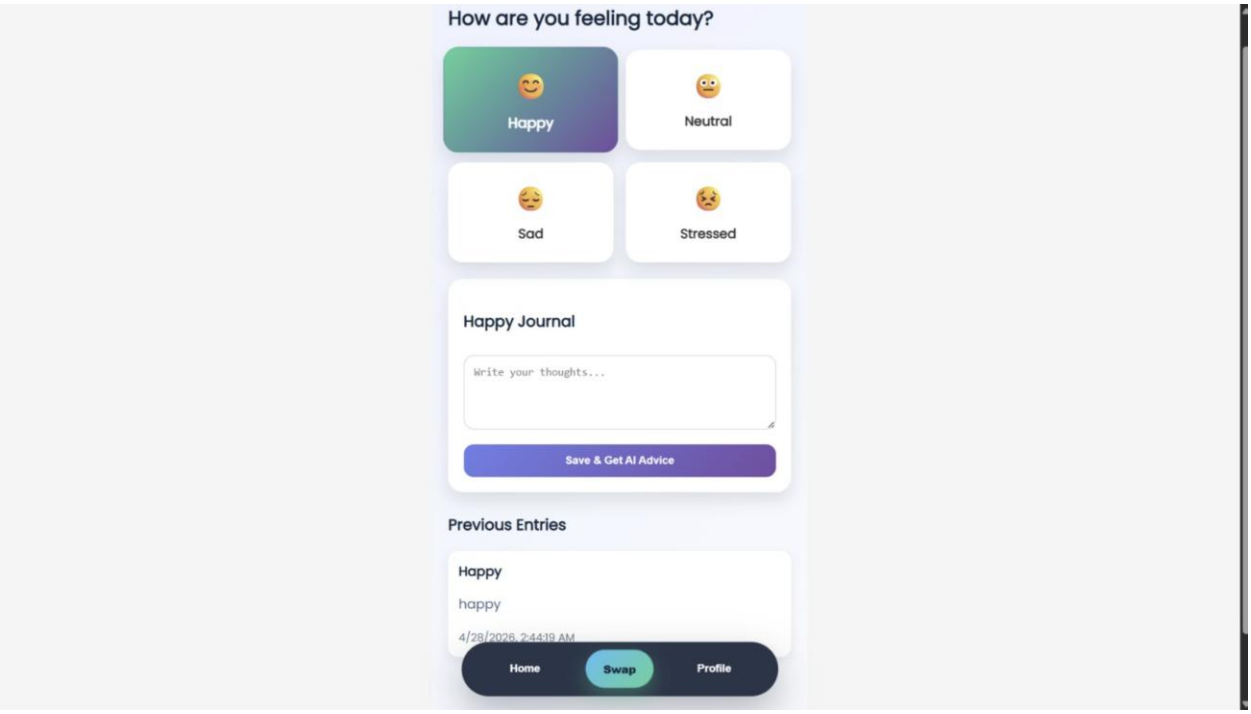
Meditation Page



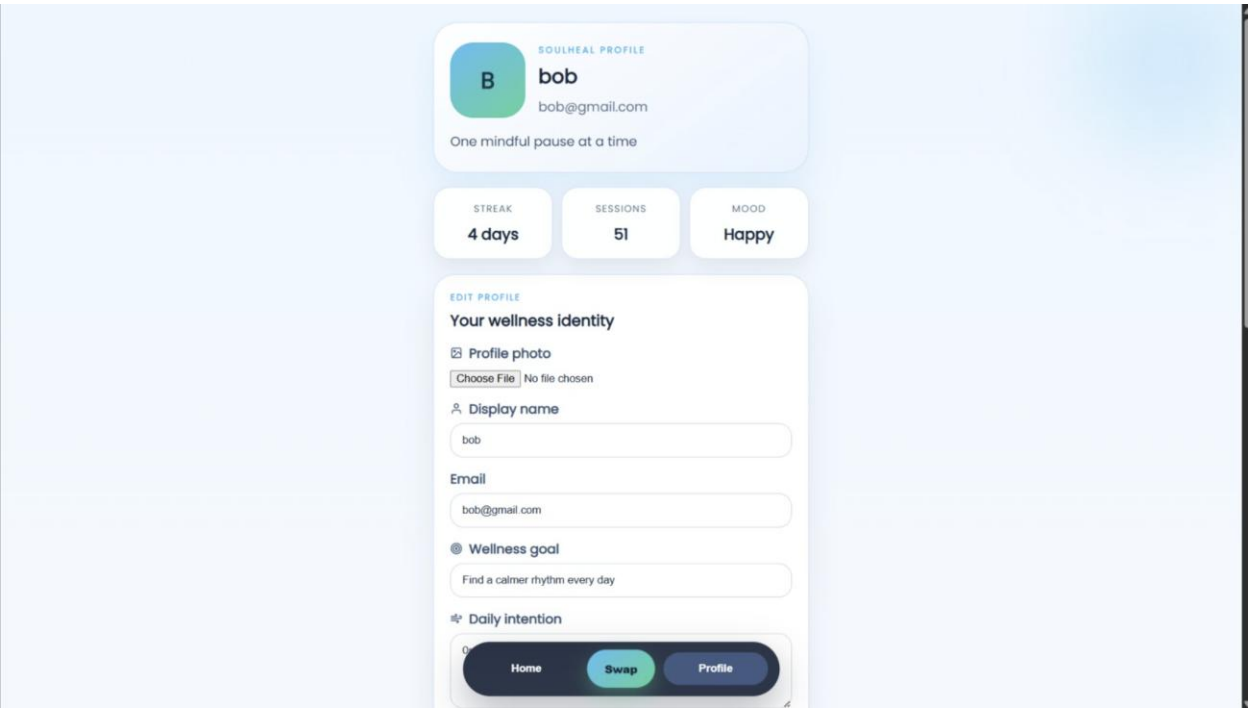
Therapy

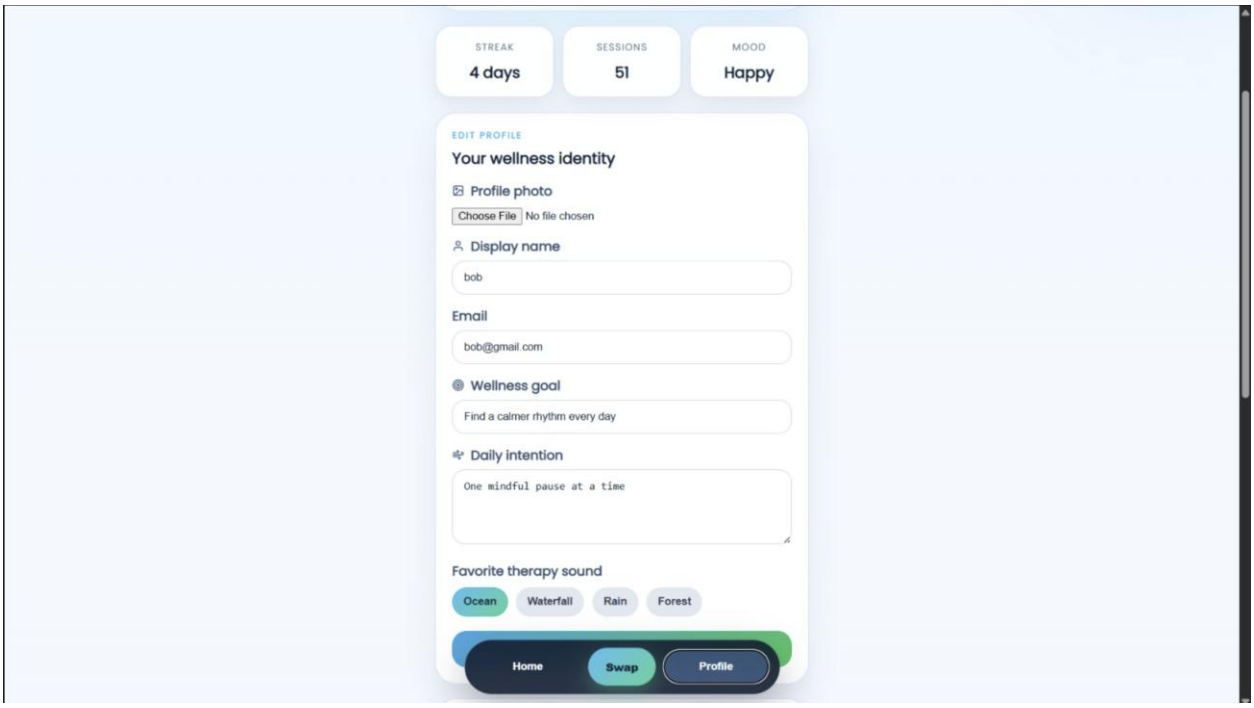


Journal:



Profile


SOULHEAL PROFILE
B
bob
bob@gmail.com
One mindful pause at a time
STREAK
4 days
SESSIONS
51
MOOD
Happy
EDIT PROFILE
Your wellness identity
Profile photo
Choose File
No file chosen
Display name
bob
Email
bob@gmail.com
Wellness goal
Find a calmer rhythm every day
Daily intention
Home
Swap
Profile


STREAK
4 days
SESSIONS
51
MOOD
Happy
EDIT PROFILE
Your wellness identity
Profile photo
Choose File
No file chosen
Display name
bob
Email
bob@gmail.com
Wellness goal
Find a calmer rhythm every day
Daily intention
One mindful pause at a time
Favorite therapy sound
Ocean
Waterfall
Rain
Forest
Home
Swap
Profile

18. Conclusion

SoulHeal addresses a critical and often overlooked need in the academic world — accessible, stigma-free mental health support for students. By combining the power of the MERN stack with thoughtful UI/UX design, the platform delivers a meaningful and practical wellness tool.

The platform demonstrates full-stack development capabilities including secure authentication, complete CRUD operations, MVC architecture, REST API design, and a polished frontend experience. Features like sound therapy, guided breathing, and mood journaling make the app genuinely useful beyond its technical scope.

SoulHeal is not just a college project — it is a real-world application that reflects how technology can be used to make mental health support more accessible, personal, and private for every student who needs it.